

SQL Project — Analysis Questions

Tables: customer · employee · order · product | Columns: customer_id · customer_name · city · region · employee_id · employee_name · department · order_id · order_date · product_id · quantity · total_amount · product_name · category · price

1 · Sales & Revenue

1	Total Revenue	SELECT SUM(total_amount) FROM order
2	Best-selling category	order JOIN product → GROUP BY category → SUM(quantity) or SUM(total_amount)
3	Monthly / yearly sales	order → FORMAT(order_date, 'YYYY-MM') + SUM(total_amount) + GROUP BY month
4	Sales Running Total	order → SUM(total_amount) OVER (ORDER BY order_date) — window function
5	Sales Moving Average (7-day)	order → AVG(total_amount) OVER (ORDER BY order_date ROWS 6 PRECEDING)
6	Sales Growth Rate vs previous month	order → LAG(monthly_sales) OVER (ORDER BY month) → (current-prev)/prev × 100
7	Avg Order Value (AOV)	order → SUM(total_amount) / COUNT(order_id)
8	Products never sold (price > \$100)	product LEFT JOIN order → WHERE order_id IS NULL AND price > 100

2 · Customer Insights

1	City with most orders	customer JOIN order → GROUP BY city → COUNT(order_id) DESC
2	Top 5 customers by revenue	customer JOIN order → GROUP BY customer_id → SUM(total_amount) DESC LIMIT 5
3	Largest region by customer base	customer → GROUP BY region → COUNT(customer_id)
4	Customers with no orders	customer LEFT JOIN order → WHERE order_id IS NULL
5	Customer Total Revenue (CLV)	customer JOIN order → SUM(total_amount) per customer_id
6	New vs returning customers	order → COUNT(DISTINCT customer_id) with first order date check per month
7	Churned customers (no order 90+ days)	order → MAX(order_date) per customer → WHERE last_order < NOW()-90

3 · Employee Performance

1	Employee with most orders handled	employee JOIN order → GROUP BY employee_id → COUNT(order_id) DESC
2	Department revenue contribution	employee JOIN order → GROUP BY department → SUM(total_amount)

3	Employee sales performance ranking	employee JOIN order → RANK() OVER (ORDER BY SUM(total_amount) DESC)
4	Monthly employee sales compare	employee JOIN order → GROUP BY employee_id, month → LAG for prior month compare

4 · Product & Inventory

1	Average product price	product → AVG(price)
2	Top 5 products by revenue	order JOIN product → GROUP BY product_id → SUM(total_amount) DESC LIMIT 5
3	Bottom 5 products (least sold)	order JOIN product → GROUP BY product_id → SUM(quantity) ASC LIMIT 5
4	Category revenue share (%)	product JOIN order → SUM per category / total × 100
5	Quantity vs total_amount correlation	order → scatter of quantity vs total_amount — check bulk discount pattern

5 · RFM Analysis (Recency · Frequency · Monetary)

R	Recency — days since last order	MAX(order_date) per customer_id → DATEDIFF(NOW(), max_order_date)
F	Frequency — total number of orders	COUNT(order_id) per customer_id
M	Monetary — total spend	SUM(total_amount) per customer_id
S	RFM Score + Segments	NTILE(5) on each dimension → Champions, At-Risk, Lost, Promising labels

6 · Cohort Analysis

1	Cohort month (first order month)	MIN(order_date) per customer_id → FORMAT as YYYY-MM cohort label
2	Month index (nth month after joining)	DATEDIFF(order_date, cohort_date) / 30 → period_number
3	Retention rate grid	COUNT(DISTINCT customer_id) per cohort × period → pivot to retention %

7 · Advanced / Bonus Questions

1	Sales compare by previous month (all employees)	order + employee → monthly pivot with LAG() window function
2	Customer Lifetime Value (CLV)	SUM(total_amount) × avg order frequency × avg customer lifespan
3	Geographic heatmap (region × revenue)	customer JOIN order → GROUP BY region → SUM(total_amount)
4	Bulk discount detection	order → total_amount / quantity per row → compare unit price vs product.price

5

High-value customer segments (Champions)

RFM → WHERE r_score=5 AND f_score=5 AND m_score=5